Initial Project Planning Template

Date	27/06/2026
Team ID	Not Assigned
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection	USN-1	As a data engineer, I can download the flood dataset from Kaggle and load it into Jupyter Notebook for analysis.	2	High	Gowthamsai Setti	24/06/2026	24/06/2026
Sprint-1	Data Visualization & Analysis	USN-2	As a data analyst, I can import all required Python libraries (NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, XGBoost, Flask, Joblib).	1	High	Karri Dhinesh	24/06/2026	24/06/2026
Sprint-1		USN-3	As a data analyst, I can read and explore the dataset to understand its structure, features (Temp, Humidity, Cloud Cover, ANNUAL, seasonal rainfall) and target variable (flood).	2	High	Karri Dhinesh	24/06/2026	24/06/2026
Sprint-1		USN-4	As a data analyst, I can perform univariate analysis — distribution plots and box plots — to study individual feature distributions and detect outliers.	2	High	Rishitha Padaga	24/06/2026	24/06/2026
Sprint-1		USN-5	As a data analyst, I can conduct multivariate analysis using a correlation heatmap and violin plots to identify relationships among features and flood occurrence.	2	High	Rishitha Padaga	24/06/2026	24/06/2026
Sprint-1		USN-6	As a data analyst, I can perform descriptive statistical analysis using head(), shape, info() and describe() to summarise key dataset insights.	1	Medium	Karri Dhinesh	24/06/2026	24/06/2026
Sprint-2	Data Pre-Processing	USN-7	As a data engineer, I can identify and handle missing values using isnull().sum() and fillna() to ensure data quality and completeness.	1	High	Sai Siddardha Bevara	25/06/2026	25/06/2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-2		USN-8	As a data engineer, I can detect and treat outliers using the IQR-based capping method to prevent extreme values from affecting model performance.	2	High	Sai Siddardha Bevara	25/06/2026	25/06/2026
Sprint-2		USN-9	As a data engineer, I can convert categorical variables into numerical representations using Label Encoding for machine learning compatibility.	1	Medium	Sai Siddardha Bevara	25/06/2026	25/06/2026
Sprint-2		USN-10	As a data engineer, I can split the dataset into independent variables (X) and target variable (y), then apply stratified train-test split (80/20) to preserve class balance.	2	High	Gowthamsai Setti	25/06/2026	25/06/2026
Sprint-2		USN-11	As a data engineer, I can apply StandardScaler feature scaling to normalise all input features and save the fitted scaler as transform.save using Joblib.	2	High	Gowthamsai Setti	25/06/2026	25/06/2026
Sprint-3	Model Building	USN-12	As an ML engineer, I can train a Decision Tree classifier and evaluate it using confusion matrix, classification report and accuracy score.	3	High	Sai Siddardha Bevara	25/06/2026	26/06/2026
Sprint-3		USN-13	As an ML engineer, I can build and test a Random Forest classifier with 200 estimators and evaluate its performance using ROC-AUC and F1-score.	3	High	Sai Siddardha Bevara	25/06/2026	26/06/2026
Sprint-3		USN-14	As an ML engineer, I can implement a KNN classifier, find the optimal K value by testing K=1 to K=20, and evaluate classification results.	3	High	Ravi Kumar Yadla	25/06/2026	26/06/2026
Sprint-3		USN-15	As an ML engineer, I can train an XGBoost classifier with scale_pos_weight to handle class imbalance and achieve 96.55% accuracy on test data.	5	High	Sai Siddardha Bevara	25/06/2026	26/06/2026
Sprint-3		USN-16	As an ML engineer, I can compare all 4 models using accuracy, F1-score and ROC-AUC metrics and select XGBoost as the best-performing model.	2	High	Ravi Kumar Yadla	26/06/2026	26/06/2026
Sprint-3		USN-17	As an ML engineer, I can save the trained XGBoost model as floods.save and the fitted StandardScaler as transform.save using Joblib for deployment.	2	High	Gowthamsai Setti	26/06/2026	26/06/2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-4	Application Building	USN-18	As a web developer, I can design and develop 4 HTML pages: home.html (landing), index.html (input form), chance.html (flood result), no_chance.html (no flood result) with dark-themed CSS styling.	5	High	Rishitha Padaga	26/06/2026	27/06/2026
Sprint-4		USN-19	As a web developer, I can build the Flask application (app.py) with 4 routes: / → home.html, /Predict → index.html, /chance → chance.html, /no_chance → no_chance.html.	5	High	Gowthamsai Setti	26/06/2026	27/06/2026
Sprint-4		USN-20	As a web developer, I can integrate the XGBoost model and StandardScaler into Flask using Joblib, load floods.save and transform.save at startup, and generate real-time flood predictions.	5	High	Gowthamsai Setti	26/06/2026	27/06/2026
Sprint-4		USN-21	As a DevOps engineer, I can create a Procfile, push the project to GitHub and deploy the Flask application on Render.com for public cloud access.	3	High	Ravi Kumar Yadla	27/06/2026	27/06/2026
Sprint-4		USN-22	As a tester, I can run and validate the web application by entering sample weather values and verifying that the prediction result, probability score and advisory actions display correctly.	2	High	Rishitha Padaga	27/06/2026	27/06/2026

Sprint Summary

Sprint	Epic	User Stories	Total Story Points	Team Lead
Sprint-1	Data Collection & EDA	USN-1 to USN-6	10	Gowthamsai Setti
Sprint-2	Data Pre-Processing	USN-7 to USN-11	8	Sai Siddardha Bevara
Sprint-3	Model Building & Evaluation	USN-12 to USN-17	18	Sai Siddardha Bevara
Sprint-4	Application Building & Deploy	USN-18 to USN-22	20	Gowthamsai Setti
TOTAL	All Epics	22 Stories	56	All Members